

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 1

CIRCLE GEOMETRY

SUGGESTED TIME: 22 PERIODS

IN THIS CHAPTER

This chapter develops the complete analytic study of the circle in the Cartesian plane: its various equations, tangents, intersections with lines and other circles, parametric form, radical axis and locus problems. It is one of the most frequently examined topics in the GCE Advanced Level Pure Mathematics with Mechanics paper.

- By the end of this chapter I can write the equation of a circle given its centre and radius, a diameter, or three points on it.
- By the end of this chapter I can convert the general equation $x^2 + y^2 + 2gx + 2fy + c = 0$ to centre-radius form and interpret g, f, c .
- By the end of this chapter I can find the equation of a tangent to a circle at a given point or from an external point.
- By the end of this chapter I can state and apply the condition for a line to touch a circle.
- By the end of this chapter I can use and convert parametric equations of a circle.
- By the end of this chapter I can compute the length of a tangent from an external point.



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the essentials to
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CHAPTER OBJECTIVES

- By the end of this chapter I can write the equation of a circle given its centre and radius, a diameter, or three points on it.
- By the end of this chapter I can convert the general equation $x^2 + y^2 + 2gx + 2fy + c = 0$ to centre-radius form and interpret g, f, c .
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- By the end of this chapter I can state and apply the condition for a line to touch a circle.
- By the end of this chapter I can use and convert parametric equations of a circle.
- By the end of this chapter I can compute the length of a tangent from an external point.
- By the end of this chapter I can find the intersections of a circle with a line or with another circle.
- By the end of this chapter I can write the equation of a circle through the intersection of two circles, or of a line and a circle, using the family-of-circles method.
- By the end of this chapter I can determine the radical axis of two circles and the conditions for two circles to touch.

- By the end of this chapter I can solve locus problems and constrained circle problems (touching axes, touching a line, centre on a line).

PREREQUISITES

- Coordinate geometry of the straight line: gradients, equations of lines, perpendicularity condition (Lower Sixth).
- Distance between two points and midpoint formula.
- Pythagoras' theorem and basic circle theorems from O-Level geometry (angle in a semicircle, perpendicular from centre to chord).
- Solving simultaneous equations, including one linear and one quadratic.
- Completing the square for a quadratic expression.

1 SECTION 1: EQUATIONS OF A CIRCLE — CENTRE, RADIUS, DIAMETER

In Yaoundé, a telecommunications engineer must position a relay antenna so that its signal covers three villages located at known points on a map: the coverage zone is a circle, and the antenna must sit at its centre. In Douala, a roundabout is designed so that a straight road just grazes it — a tangent problem. When a water tank's circular rim must pass through three given pillars, or when two circular irrigation zones overlap, every question reduces to finding the equation of a circle and studying its position relative to lines and other circles. This chapter gives you the algebraic tools to solve all these problems with precision.

1.1 The Circle as a Locus — Definition and the Origin-Centred Circle

DEFINITION Circle as a Locus

A **circle** is the set of all points in a plane that are at a fixed distance, called the **radius**, from a fixed point called the **centre**.

Let the centre be the origin $O(0, 0)$ and the radius be $r > 0$. For any point $P(x, y)$ on the circle, the distance OP equals r . By the distance formula:

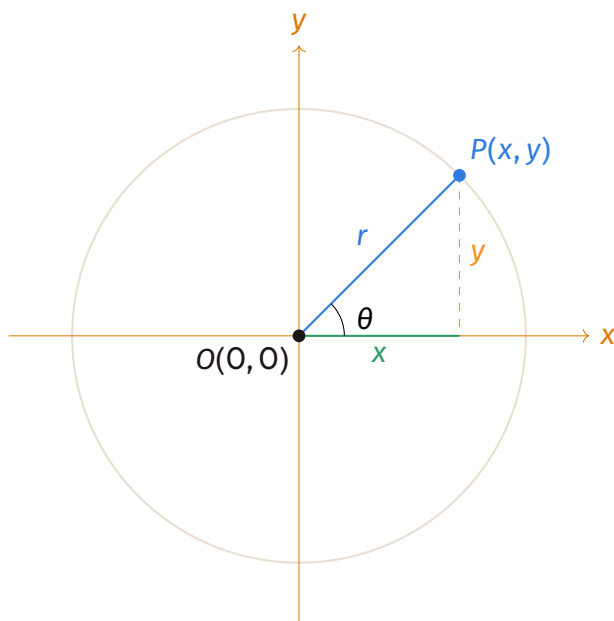
$$OP = \sqrt{(x - 0)^2 + (y - 0)^2} = \sqrt{x^2 + y^2}.$$

Hence $\sqrt{x^2 + y^2} = r$, and squaring both sides (both are non-negative, so squaring preserves equality) gives the fundamental equation.

PROPERTY Circle Centred at the Origin

The equation of a circle with centre $O(0, 0)$ and radius r is

$$x^2 + y^2 = r^2.$$



Circle centred at the origin: right triangle with legs $|x|$, $|y|$ and hypotenuse r , so $x^2 + y^2 = r^2$ by Pythagoras.

EXAMPLE Reading the radius from the equation

Find the radius of the circle $x^2 + y^2 = 50$.

$$r^2 = 50 \Rightarrow r = \sqrt{50} = 5\sqrt{2}.$$

EXAMPLE Writing the equation given the radius

Write the equation of the circle centred at the origin with radius $\sqrt{13}$.

$$x^2 + y^2 = (\sqrt{13})^2 = 13.$$

1.2 Translation of the Centre — Equation $(x - a)^2 + (y - b)^2 = r^2$

When the centre is $C(a, b)$, the distance CP for $P(x, y)$ is

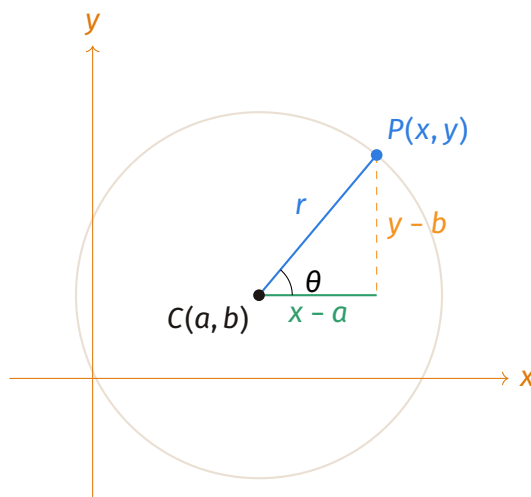
$$CP = \sqrt{(x - a)^2 + (y - b)^2}.$$

Setting $CP = r$ and squaring yields the standard form.

PROPERTY Standard Equation of a Circle

The circle with centre $C(a, b)$ and radius r has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$



Circle with centre $C(a, b)$: the right triangle has legs $x - a$ and $y - b$, so $(x - a)^2 + (y - b)^2 = r^2$.

KEY POINTS Reading Centre and Radius

From $(x - a)^2 + (y - b)^2 = r^2$:

- Centre: $C(a, b)$.
- Radius: $r = \sqrt{r^2}$ (positive square root).

Watch the signs: $(x - 3)^2$ means $a = 3$; $(x + 2)^2 = (x - (-2))^2$ means $a = -2$.

WARNING Sign Trap

In $(x - a)^2 + (y - b)^2 = r^2$ the centre is (a, b) , not $(-a, -b)$. Example: $(x + 4)^2 + (y - 5)^2 = 9$ has centre $(-4, 5)$, radius 3.

WORKED EXAMPLE 1 Centre and radius from the equation

Find the centre and radius of the circle $(x - 2)^2 + (y + 3)^2 = 25$.

SOLUTION

Comparing with $(x - a)^2 + (y - b)^2 = r^2$: $a = 2, b = -3, r^2 = 25 \Rightarrow r = 5$. Centre $C(2, -3)$, radius 5.

WORKED EXAMPLE 2 Writing the equation given centre and radius

Find the equation of the circle with centre $C(-1, 4)$ and radius 6.

SOLUTION

$$a = -1, b = 4, r = 6 \Rightarrow r^2 = 36.$$

$$(x - (-1))^2 + (y - 4)^2 = 36 \Rightarrow (x + 1)^2 + (y - 4)^2 = 36.$$

1.3 Circle Given Centre and a Point on the Circle (Lesson 3)

If the centre $C(a, b)$ is known and a point $P(x_1, y_1)$ lies on the circle, the radius is the distance CP :

$$r = CP = \sqrt{(x_1 - a)^2 + (y_1 - b)^2}.$$

Then the equation is $(x - a)^2 + (y - b)^2 = r^2$.

METHOD Circle from Centre and a Point

1. Compute $r^2 = (x_1 - a)^2 + (y_1 - b)^2$.
2. Write $(x - a)^2 + (y - b)^2 = r^2$.

WORKED EXAMPLE 3 Centre and a point on the circle

Find the equation of the circle with centre $C(3, -2)$ passing through $P(6, 2)$.

SOLUTION

$$r^2 = (6 - 3)^2 + (2 - (-2))^2 = 3^2 + 4^2 = 9 + 16 = 25. \text{ Equation: } (x - 3)^2 + (y + 2)^2 = 25.$$

1.4 Circle on a Given Diameter (Lesson 5)

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the endpoints of a diameter.

PROPERTY Centre and Radius from Diameter

- Centre M is the midpoint of AB :

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

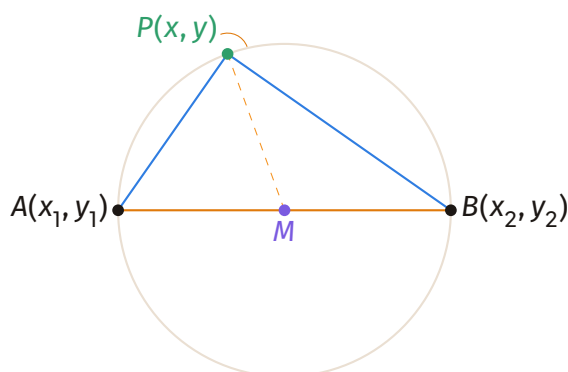
- Radius $r = \frac{1}{2}AB = \frac{1}{2}\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

PROPERTY Alternative Form — Angle in a Semicircle

For any point $P(x, y)$ on the circle, $\angle APB = 90^\circ$ (angle in a semicircle), so $\vec{PA} \cdot \vec{PB} = 0$:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0.$$

This is the **diameter form** of the circle equation.



Circle on diameter AB : $\angle APB = 90^\circ$ gives $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.

METHOD Circle on Diameter AB

- Midpoint method: Find $M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$, compute $r = \frac{1}{2}AB$, write $(x - x_M)^2 + (y - y_M)^2 = r^2$.
- Diameter form (often faster): Write directly $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ and expand if needed.

WORKED EXAMPLE 4 Circle on a diameter

Find the equation of the circle on $A(2, -1)$ and $B(6, 3)$ as diameter.

SOLUTION

Method 1 (Midpoint): $M\left(\frac{2+6}{2}, \frac{-1+3}{2}\right) = (4, 1)$. $AB = \sqrt{(6-2)^2 + (3-(-1))^2} = \sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$.
 $r = 2\sqrt{2}$, $r^2 = 8$. Equation: $(x-4)^2 + (y-1)^2 = 8$.

Method 2 (Diameter form): $(x-2)(x-6) + (y+1)(y-3) = 0$. $x^2 - 8x + 12 + y^2 - 2y - 3 = 0 \Rightarrow x^2 + y^2 - 8x - 2y + 9 = 0$. Complete squares: $(x-4)^2 - 16 + (y-1)^2 - 1 + 9 = 0 \Rightarrow (x-4)^2 + (y-1)^2 = 8$.

**1.5 Position of a Point Relative to a Circle****PROPERTY** Point Inside / On / Outside

Let circle C have centre $O(a, b)$ and radius r . For a point $P(x_0, y_0)$, compute $d = OP = \sqrt{(x_0 - a)^2 + (y_0 - b)^2}$.

- $d < r \Rightarrow P$ is **inside** the circle.
- $d = r \Rightarrow P$ lies **on** the circle.
- $d > r \Rightarrow P$ is **outside** the circle.

Equivalently, compare $(x_0 - a)^2 + (y_0 - b)^2$ with r^2 (avoids the square root).

WORKED EXAMPLE 5 Position of a point

Determine whether $P(5, 2)$ lies inside, on, or outside the circle $(x-1)^2 + (y+2)^2 = 20$.

SOLUTION

Centre $C(1, -2)$, $r^2 = 20$. $CP^2 = (5-1)^2 + (2-(-2))^2 = 4^2 + 4^2 = 16 + 16 = 32$. Since $32 > 20$, $CP^2 > r^2$, so P is outside the circle.

EXERCISE 1 Practice 1.1

1. Write the equation of the circle centred at the origin with radius $\sqrt{17}$.
2. Find the centre and radius of $(x+3)^2 + (y-4)^2 = 49$.

3. Find the equation of the circle with centre $(-2, 5)$ passing through $(1, 9)$.
4. Find the equation of the circle on $A(-3, 2)$ and $B(5, -4)$ as diameter (use both methods).
5. Determine the position of $Q(0, 0)$ relative to the circle $(x - 2)^2 + (y + 3)^2 = 10$.

SOLUTION Practice 1.1

1. $x^2 + y^2 = 17$.
2. Centre $(-3, 4)$, radius 7.
3. $r^2 = (1 + 2)^2 + (9 - 5)^2 = 9 + 16 = 25$; $(x + 2)^2 + (y - 5)^2 = 25$.
4. Midpoint $M(1, -1)$, $AB = \sqrt{8^2 + (-6)^2} = 10$, $r = 5$; $(x - 1)^2 + (y + 1)^2 = 25$. Diameter form:
 $(x + 3)(x - 5) + (y - 2)(y + 4) = 0 \Rightarrow x^2 - 2x - 15 + y^2 + 2y - 8 = 0 \Rightarrow (x - 1)^2 + (y + 1)^2 = 25$.
5. $C(2, -3)$, $r^2 = 10$. $CQ^2 = (0 - 2)^2 + (0 + 3)^2 = 4 + 9 = 13 > 10$; Q is outside.

DID YOU KNOW? Historical Note

The Greek mathematician Apollonius of Perga (c. 262–190 BC) systematically studied conic sections, including the circle. The coordinate approach we use today was developed by René Descartes (1596–1650) and Pierre de Fermat (1607–1665), linking geometry and algebra — the foundation of analytic geometry taught in Cameroonian lycées since the 1960s.

2 SECTION 2: THE GENERAL EQUATION AND CIRCLES THROUGH THREE POINTS

In Section 1 we built the equation of a circle from its geometric definition: the set of points at a fixed distance r from a fixed centre (a, b) . This gave us the **centre–radius form**

$$(x - a)^2 + (y - b)^2 = r^2.$$

This form is ideal when you know the centre and radius. But in many GCE problems the equation arrives already expanded, looking like $x^2 + y^2 - 6x + 4y - 12 = 0$, and nothing about it immediately reveals the centre or radius. Worse, sometimes you are not given the centre at all — only three points the circle must pass through. This section gives you the two essential tools: how to recognise and decode an expanded circle equation (the **general form**), and how to construct a circle from three non-collinear points.

2.1 From Centre–Radius Form to the General Equation

Start from the centre–radius form and simply expand the brackets:

$$\begin{aligned}(x - a)^2 + (y - b)^2 &= r^2 \\ x^2 - 2ax + a^2 + y^2 - 2by + b^2 &= r^2 \\ x^2 + y^2 - 2ax - 2by + (a^2 + b^2 - r^2) &= 0.\end{aligned}$$

Now rename the constants: put $g = -a$, $f = -b$ and $c = a^2 + b^2 - r^2$. The equation becomes the famous **general equation of a circle**.

DEFINITION General equation of a circle

The **general equation** of a circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0,$$

where g, f, c are real constants. Its **centre** is $(-g, -f)$ and its **radius** is

$$r = \sqrt{g^2 + f^2 - c}.$$

Notice the bookkeeping: the centre coordinates are minus half the coefficients of x and y (because the coefficients are $2g$ and $2f$), and the constant term c absorbs $a^2 + b^2 - r^2$.

METHOD Completing the square: general form $\rightarrow$ centre–radius form

Given $x^2 + y^2 + 2gx + 2fy + c = 0$:

1. Group the x -terms and the y -terms: $(x^2 + 2gx) + (y^2 + 2fy) = -c$.
2. Complete each square: add g^2 and f^2 to both sides:

$$(x + g)^2 - g^2 + (y + f)^2 - f^2 = -c.$$

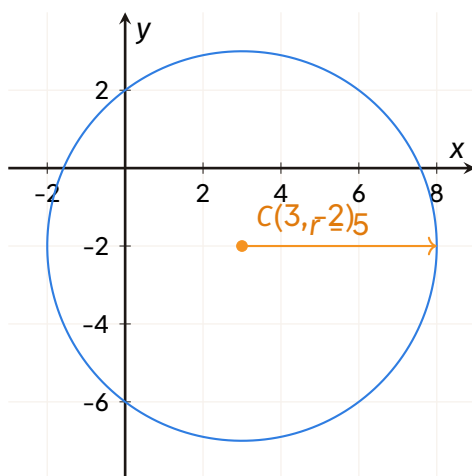
3. Rearrange: $(x + g)^2 + (y + f)^2 = g^2 + f^2 - c$.
4. Read off: centre $(-g, -f)$, radius $\sqrt{g^2 + f^2 - c}$.

EXAMPLE Decoding a general equation

Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

By formula: $2g = -6 \Rightarrow g = -3$; $2f = 4 \Rightarrow f = 2$; $c = -12$. Centre $= (-g, -f) = (3, -2)$, radius $= \sqrt{9 + 4 + 12} = \sqrt{25} = 5$.

By completing the square: $(x^2 - 6x) + (y^2 + 4y) = 12$, so $(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$, i.e. $(x - 3)^2 + (y + 2)^2 = 25$. Same answer — and this is exactly the circle of Section 1.



The circle $x^2 + y^2 - 6x + 4y - 12 = 0$: centre $(-g, -f) = (3, -2)$, radius $\sqrt{g^2 + f^2 - c} = 5$.

2.2 Does the General Equation Always Give a Circle?

WARNING When is it really a circle?

The radius formula involves $\sqrt{g^2 + f^2 - c}$, so we must examine the sign of $g^2 + f^2 - c$:

- $g^2 + f^2 - c > 0$: a genuine circle of positive radius;
- $g^2 + f^2 - c = 0$: a **point circle** — the locus is the single point $(-g, -f)$;
- $g^2 + f^2 - c < 0$: no real locus at all (a sum of two squares cannot be negative).

For example $x^2 + y^2 + 2x + 4y + 9 = 0$ gives $g^2 + f^2 - c = 1 + 4 - 9 = -4 < 0$: no real point satisfies it.

WARNING Not every quadratic is a circle

An equation represents a circle only if

1. the coefficients of x^2 and y^2 are equal (and non-zero), and
2. there is no xy term.

Thus $2x^2 + 2y^2 + 3x - 4y + 1 = 0$ is a circle (divide by 2 first!), but $x^2 + 2y^2 - 4x = 0$ is an ellipse and $x^2 + y^2 + 3xy = 1$ is not a circle either. Always divide through to make the coefficients of x^2 and y^2 equal to 1 before reading off g, f, c .